

# **On the Dynamics, Equilibria, and Economics of Gender Balance at Naturist Venues**

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## Table of Contents

|                                                                  |    |
|------------------------------------------------------------------|----|
| Abstract.....                                                    |    |
| Differential Model of Naturist Venue Demographics.....           | 1  |
| Stability of Phase-Space Solutions.....                          | 5  |
| A compact proof of stability <sup>1</sup> .....                  | 9  |
| The effects of male indifference to male predominance .....      | 10 |
| Regarding Closed-form Solutions in the Time Domain.....          | 12 |
| Forcing Viability: the “No Single Males” policy.....             | 14 |
| Shadow Costs: Unlocking the Value of Others.....                 | 16 |
| Scenario A: Maximisation of Revenue ( $Z$ ).....                 | 17 |
| Scenario B: Maximisation of Attendance ( $W$ ).....              | 19 |
| Trading Off: Marginal Rates of Substitution.....                 | 20 |
| Full House: Finite Carrying Capacity.....                        | 22 |
| Dynamics at Capacity: the “Churn Mechanism” .....                | 23 |
| When women are at ease.....                                      | 25 |
| The Geometry of Safety .....                                     | 28 |
| Conclusion.....                                                  | 30 |
| On the “no single male” policy: motivation and consequences..... | 32 |
| Stepping back.....                                               | 34 |
| Reference List.....                                              | 35 |
| Appendix.....                                                    | 36 |

## Abstract

This paper proposes a dynamical systems approach to modelling the demographic evolution of naturist venues. By quantifying gender-specific sensitivities to gender ratios, we derive an equation that governs the evolution of the system’s demographic trajectory. Stability analysis reveals a fundamental asymmetry: the “invisible hand” does not guarantee equilibrium, but rather admits a critical “tipping point” below which the venue collapses into a male-dominated state. We further analyse the “no single male” policy as a binding mathematical constraint, using standard KKT conditions to derive the “shadow price” of female attendance and demonstrating that cross-subsidation strategies are rational tools for attendance- and revenue-maximisation. Finally, we apply Voronoi tessellation and percolation theory to contrast administrative constraints with intrinsic stability, demonstrating that a high-trust community configuration allows for organic growth that is inherently robust to gender-balance breaking shocks, rendering strict administrative constraints redundant in mature communities. In the conclusion, we invite the reader to reflect on whether the “no single male” policy is a necessary shield or merely a remedy for insufficient community cohesion.

On the Dynamics, Equilibria, and Economics of Gender Balance at Naturist Venues  
Differential Model of Naturist Venue Demographics

To our knowledge, little effort has been put into quantifying or even characterising the implications and consequences of oft anecdotally discussed gender balance, or rather imbalance, observed in naturist venues. To this end, we propose a simple system of ordinary differential equations (ODEs) to model the evolution of demographics at such venues based on fixed coefficients quantifying pools of potential participants' attitudes towards the ratio of genders.

Given  $f$  females and  $m$  males, over time  $t$  evolution is posited to obey

$$\begin{cases} \frac{df(t)}{dt} = \varphi_1 \cdot \frac{f(t)}{m(t)} - \varphi_2 \cdot \frac{m(t)}{f(t)} \\ \frac{dm(t)}{dt} = \mu_1 \cdot \frac{f(t)}{m(t)} - \mu_2 \cdot \frac{m(t)}{f(t)} \end{cases} \quad (1)$$

where  $\varphi_1$  is females' preference for a higher female-to-male ratio  $f/m$ ,  $\varphi_2$  is females' dissatisfaction with a higher male-to-female ratio  $m/f$ ,  $\mu_1$  is males' preference for a higher female-to-male ratio  $f/m$ , and  $\mu_2$  is males' dissatisfaction with a higher male-to-female ratio  $m/f$ . It is assumed, at least initially, that  $\varphi_1, \varphi_2, \mu_1, \mu_2 > 0$  and that  $\varphi_1, \varphi_2, \mu_1, \mu_2 \in \mathbb{R}$ .

In some sense the system can be interpreted as an interplay between:

- (a) **“Comfort terms”**  $f(t)/m(t)$ , where a higher ratio of females to males generally increases the ‘comfort’ and perceived “social safety” for everyone. For females, it reduces the feeling of being stared at or being the object of unwanted attention; for males, it validates the venue as a popular, balanced social environment; and
- (b) **“Pressure terms”**  $m(t)/f(t)$ , where a higher ratio of males to females creates a ‘skewed’ environment. In naturist venues, a high male-to-female ratio is often perceived as ‘creepy’ or ‘unbalanced’, leading to a decrease in the growth rate of both populations (though sensitivity can vary markedly, as discussed later).

It is assumed that *a priori* no inherent gender bias in favour or against practicing

naturism exists but that observed rates of participation in public venues are driven by the gender ratios themselves. We will find that females' displeasure at being in considerable minority and males' relative indifference to high proportions of their own gender come to dominate the behaviour of the system. To state the obvious, as female proportion increases, both populations experience growth (or more precisely, an influx of participants), while as male proportion increases, both populations experience decline (or efflux).

The objective of such a toy model is to capture the essence of the phenomenon while remaining broadly tractable analytically and thus serving as a source of insight, rather than quantitative accuracy that would probably be better attained by other means. Nonetheless we shall find that such a simplistic model (further extended by including finite capacity) has considerable descriptive power and captures many core features observed anecdotally in the field.

In the interests of brevity and clarity the dependence of  $f$  and  $m$  on  $t$  will be henceforth dropped from discussions of (1), leading to a restatement in condensed form:

$$\begin{cases} \dot{f} = \varphi_1 \cdot \frac{f}{m} - \varphi_2 \cdot \frac{m}{f} \\ \dot{m} = \mu_1 \cdot \frac{f}{m} - \mu_2 \cdot \frac{m}{f} \end{cases} \quad \text{with } \varphi_i, \mu_i, \in \mathbb{R}_{>0}. \quad (2)$$

To solve for the phase-space population trajectories, we first introduce the ratio function  $r \doteq f/m > 0$  such that  $f = xm$ . Since the system is homogenous and of degree zero, the phase trajectories are determined by the relationship between  $f$  and  $m$ .

Using the quotient rule, we can take the derivative of  $r$ :

$$\dot{r} = \frac{\dot{f}m - f\dot{m}}{m^2} = \frac{\dot{f}}{m} - \frac{f}{m} \cdot \frac{\dot{m}}{m}. \quad (3)$$

Substituting the expressions for  $\dot{f}$  and  $\dot{m}$  gives:

$$\dot{r} = \frac{1}{m} \left( \varphi_1 - \frac{\varphi_2}{r} \right) - \frac{r}{m} \left( \mu_1 r - \frac{\mu_2}{r} \right). \quad (4)$$

Factoring out  $1/m$  and distributing  $r$  gives us

$$\dot{r} = \frac{1}{m} \left( -\mu_1 r^2 + \varphi_1 r + \mu_2 - \frac{\varphi_2}{r} \right). \quad (5)$$

To find trajectories in the  $(m, f)$  plane we eliminate time  $t$  by taking the ratio of the two original ODEs and then simplify by multiplying both numerator and denominator by the term  $f/m$ , and finally using  $r = f/m$ :

$$\frac{df}{dm} = \frac{\dot{f}}{\dot{m}} = \frac{\varphi_1 \frac{f}{m} - \varphi_2 \frac{m}{f}}{\mu_1 \frac{f}{m} - \mu_2 \frac{m}{f}} = \frac{\varphi_1 (\frac{f}{m})^2 - \varphi_2}{\mu_1 (\frac{f}{m})^2 - \mu_2} = \frac{\varphi_1 r^2 - \varphi_2}{\mu_1 r^2 - \mu_2}. \quad (6)$$

Now we isolate the variables  $m$  and  $r$  to perform integration:

$$m \frac{dr}{dm} = \frac{\varphi_1 r^2 - \varphi_2}{\mu_1 r^2 - \mu_2} - r = \frac{\varphi_1 r^2 - \varphi_2 - \mu_1 r^3 + \mu_2 r}{\mu_1 r^2 - \mu_2}. \quad (7)$$

Separating the variables gives us:

$$\frac{\mu_1 r^2 - \mu_2}{-\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2} dr = \frac{dm}{m}. \quad (8)$$

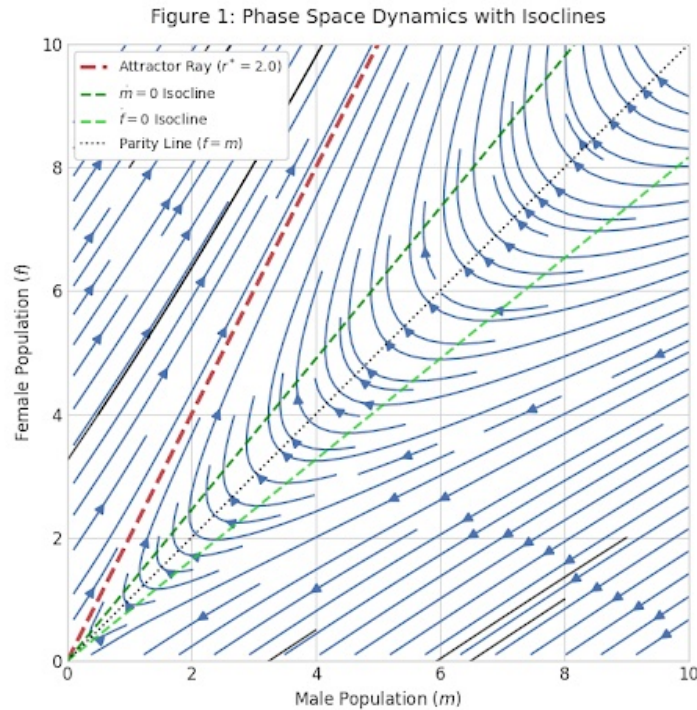
The integral on the left depends on the roots of the cubic polynomial

$$P(r) \doteq r \cdot \left( \mu_2 + \varphi_1 r - \mu_1 r^2 - \frac{\varphi_2}{r} \right) = -\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2, \quad (9)$$

and two cases are worth distinguishing:

- (a) **Fixed Rays** when  $P(r) = 0$  has a positive real root  $r^*$  then  $dm/dr = 0$  which represents a straight-line trajectory in phase-space where  $f = r^* m$  that function as an attractor to which general curves tend to or away from, but cannot cross; and
- (b) **General Curves** where  $\dot{r} \neq 0$  the trajectories will curve towards or away from those fixed rays, defining the flow of the naturist venue's population levels.

In the sample diagram, the streamlines governing the system's behaviour are depicted in blue. The red dashed line is a fixed ray attractor, and in this example since it is below the  $45^\circ$  incline it shows circumstances where males (slightly) outweigh females. The green dashed line represents the isocline of zero female growth  $\dot{f} = 0$  where the number of females joining exactly matches the number of females leaving. Below that, the shaded area represents a region where female population collapses to zero. A relatively high tolerance for predominantly male demographics ( $\mu_2 \approx 0$ ) ensures the remaining population is all male.



**The one-line take-away is that in this system of ODEs modelling gender balance at naturist venues *equilibria are ratios of relative populations and not specific values.***

### Stability of Phase-Space Solutions

Using the quotient rule on  $r = f/m$  gives

$$\dot{r} = \frac{\dot{f}m - \dot{m}f}{m^2} = \frac{1}{m}(\dot{f} - r\dot{m}) \quad (10)$$

into which we can substitute our system equations, yielding

$$\dot{r} = \frac{1}{m} \left[ \left( \varphi_1 r - \frac{\varphi_2}{r} \right) - r \left( \mu_1 r - \frac{\mu_2}{r} \right) \right]. \quad (11)$$

Combining denominators by multiplying within the bracket by  $r/r$  gives us the equation that governs the overall evolution of the gender ratio:

$$\dot{r} = \frac{1}{mr} (-\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2) = \frac{1}{mr} \cdot P(r). \quad (12)$$

Since the population  $m$  and the ratio  $r$  are strictly positive quantities, the sign of the rate of change of  $\dot{r}$  depends entirely on the characteristic polynomial  $P(r)$ . The equilibrium points of the system are the roots  $r^*$  where

$$P(r^*) = 0. \quad (13)$$

Not all equilibria are stable, however. To determine long-term fate of the venue, we analyse the stability of these roots by examining the derivative of the growth-rate  $d\dot{r}/dr$  at the equilibrium point itself. Using the chain rule on the equation that governs the overall evolution of the gender ratio (11):

$$\left. \frac{d\dot{r}}{dr} \right|_{r=r^*} = \frac{\frac{dP}{dr}(r^*)}{mr^*}. \quad (14)$$

Since the prefactor  $1/(mr^*)$  is always positive, the stability is determined solely by the slope of the polynomial  $P(r^*)$  as it crosses the axis:

If  $P(r^*) < 0$  this  $r^*$  is a stable attractor because the polynomial slopes downwards towards this root, ensuring that any deviation will be self-correcting. This represents the venue's "social equilibrium" ratio. It is the value around which the venue's gender balance will naturally cluster.



If  $P(r^*) > 0$  this  $r^*$  is an unstable repulsor because as the polynomial slopes upwards, any deviation will be self-reinforcing. This can be thought of as representing a “critical ratio”. If the ratio falls beneath this critical value, the self-reinforcing feedback loop takes hold and accelerates until the female proportion vanishes.

In general, we have three distinct demographic behaviours:

If  $P(r) > 0$  then the proportion of females increases;

If  $P(r) = 0$  the system is in equilibrium and the relative proportions remain constant;

If  $P(r) < 0$  the proportion of males increases.

Let us investigate the extrema of  $P(r)$  in order to characterise its behaviour and thus perform boundary analysis keeping  $m, f > 0$  because the model is undefined on the axes.

At  $P(0) = -\varphi_2$  because all other terms are factors of  $r$ . Since  $\varphi_2$  (female exit rate) is positive,  $P(0)$  must be negative. Demographically this implies the if the ratio is very low, the *change in the rate*  $\dot{r}$  is negative, and the system will be driven towards total collapse ( $r = 0$ ) and there is no mechanism of endogenous recovery from this final population of zero.

At  $r \rightarrow \infty$  (infinite females) the leading term  $-\mu_1 r^3$  dominates. Since  $\mu_1$  (male influx) is positive,  $P(r) \rightarrow -\infty$ . The demographic interpretation is that if the female population were to become overwhelmingly predominant somehow, a correspondingly oversized male influx will inevitably bring the ratio back down.

As the curve defined by  $P(r)$  begins negative at  $r = 0$  and goes to negative infinity at arbitrarily large  $r$ , there are three cases, of which two scenarios are likely for a venue:

No positive roots imply a “doomed venue” where the coefficients of  $P(r)$  are such that its local maximum remains below zero, and thus  $\dot{r}$  is always negative. Demographically this entails that irrespective of how many females attend initially, the ratio will relentlessly decrease until the female population vanishes.

Two positive roots imply a “viable venue” where the attraction parameters  $(\varphi_1, \mu_2)$  are

strong enough relative to the “exit/entry” pressures to make the cubic curve bulge upwards and cross the axis twice at roots that we shall denote  $r_1$  and  $r_2$ . We can characterise these roots on the basis of the gradient of  $P(r)$ .

The repeated double root corner-case implies a “dicey venue” because  $P(r) \approx -a(r - r^*)^2$  (where  $a$  is the tangent at the local maximum) and thus  $P(r) \leq 0$  on both sides of  $r^*$  (strictly negative except at the root). In reality we do not expect to observe such a scenario as it is so unstable to likely be merely transient.

The lower root  $r_1$  is a “viability threshold” where the curve makes its first crossing from negative to positive territory. Its slope is positive and this is an *unstable repulsor*, and the dynamics are as follows:

If the system begins slightly below this root ( $r < r_1$ ),  $P(r)$  is negative and  $\dot{r} < 0$  so the system will be pushed down towards zero (collapse of female population).

If the system begins slightly above this root ( $r > r_1$ ),  $P(r)$  is positive and  $\dot{r} > 0$  so the system will be pushed upwards towards the second root. This is the “critical ratio” the venue must stay above in order to remain viable.

The upper root  $r_2$  is a “natural equilibrium” where the curve would make a second crossing from positive territory back into negative values. The slope here is negative and it is a *stable attractor*: if the system is above this value,  $\dot{r} < 0$  (the ratio increases), and if the system is below this value,  $\dot{r} > 0$  (the ratio decreases). This represents the long-term destiny of the venue: as long as the ratio stays above this threshold, the demographics will naturally settle here.

Summarising the stability analysis, the general condition for a sustainable naturist venue in this model is that the polynomial  $P(r)$  must have a local maximum greater than zero. We can find the local maximum by solving

$$P'(r) = -3\mu_1 r^2 + 2\varphi_1 r + \mu_2 = 0. \quad (15)$$

If the value of  $P(r)$  at this maximum is positive, the venue will survive and maybe thrive. If it is negative, the resort is destined to fail.

A compact proof of stability<sup>1</sup>

Because the system is homogenous of degree zero, stability of an equilibrium ray  $r^*$  is governed by the scalar

$$\eta(r_2^*) = \left. \frac{d\dot{r}}{dr} \right|_{r=r_2^*} = P'(r_2^*) < 0. \quad (16)$$

This is geometrically equivalent to requiring that the flow of the vector field across the ray is directed inwards.

When  $\varphi_2 = 0$ , the cubic polynomial (9) loses its constant term and can be factored as  $P(r) = r \cdot (-\mu_1 r^2 + \varphi_1 r + \mu_2)$ , so immediately  $P(0) = 0$ ; furthermore when  $\varphi_2 = \mu_2 = 0$ , (9) can be further factored as  $P(r) = r^2 \cdot (-\mu_1 r + \varphi_1)$ : intuitively enough, if efflux coefficients vanish, stable growth dominates as without sensitivity to pressure terms, there is nothing holding participants back.

Conversely, if the trajectory slope is steeper than the ray when above it, the gap widens.

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1. Sophisticated readers will note that this is essentially an eigenvalue argument without the need to roll out the vector-calculus heavy artillery.

*The effects of male indifference to male predominance*

Intuition guides into believing that males have a relatively high tolerance for predominantly male demographics  $\mu_2 \approx 0$ . We can harness this intuition and make males totally indifferent by setting  $\mu_2 = 0$ . This represents a resort where males do not depart even if females are vastly outnumbered. This changes the dynamics significantly: without the “pressure relief valve” of males departing, the venue is much more prone to collapse if the female population diminishes.

When  $\mu_2 = 0$ , the governing polynomial loses its linear term:  $P(r) = -\mu_1 r^3 + \varphi_1 r^2 - \varphi_2$ , creating a very specific “geometry of viability”. The function has a local maximum at  $r_{max} = \frac{3\mu_1}{2\varphi_1}$ , and in order for this venue to continue being viable, this peak must be positive ( $P(r_{max}) > 0$ ), leading to the inequality

$$P(r_{max}) > 0 \iff \frac{4\varphi_1^3}{27\mu_1^2} - \varphi_2 > 0 \iff \varphi_1^3 > \frac{27}{4}\mu_1^2\varphi_2 \quad (17)$$

whose demographic interpretation is that the influx of females  $\varphi_1$  relative to that of males  $\mu_1$  must be great enough to overcome the basic female efflux rate  $\varphi_2$ . If this condition is satisfied, we have three roots for  $P(r)$ :

An unphysical negative root  $\rho_1$  that can be ignored;

An unstable repulsor threshold root  $\rho_2$  beneath which ratio the venue is destined for collapse;

A stable attractor root  $\rho_3$  at which the venue’s demographic ratio will naturally settle.

We are able to solve for the population trajectories in this scenario. From (8) we know that

$$\frac{dm}{m} = \frac{\mu_1 r^2}{-\mu_1(r - \rho_1)(r - \rho_2)(r - \rho_3)} dr. \quad (18)$$

Cancelling  $\mu_1$  and using partial fraction decomposition we have:

$$\frac{r^2}{(r - \rho_1)(r - \rho_2)(r - \rho_3)} = \frac{k_1}{r - \rho_1} + \frac{k_2}{r - \rho_2} + \frac{k_3}{r - \rho_3} \quad (19)$$

Integrating both sides gives us the exact shape of the curves in the phase portrait:

$$m(r) = C_0 \cdot |r - \rho_1|^{-k_1} \cdot |r - \rho_2|^{-k_2} \cdot |r - \rho_3|^{-k_3}. \quad (20)$$

(where coefficients  $k_i$  are derived from roots in the manner of  $k_1 = \frac{\rho_1^2}{(\rho_1 - \rho_2)(\rho_1 - \rho_3)}$  & *cetera*).

We are remarkably also in a position to calculate  $t(r)$ : how long it takes a venue to reach a gender ratio (including the special case  $t(0)$ , the moment of collapse). We do this by substituting  $m(r)$  into the male rate equation  $\dot{r} = \frac{dm}{dt} = \mu_1 r$  to obtain:

$$t(r) = \frac{1}{\mu_1} \int_{r_0}^r \frac{1}{s \cdot m(s)} dm. \quad (21)$$

Since that  $dm = m \frac{\mu_1 s^2}{P(s)} ds$  we can rewrite this as

$$t(r) = \int_{r_0}^r \frac{m(s) \cdot s}{P(s)} ds. \quad (22)$$

For illustrative purposes, consider the following “death spiral” scenario, wherein a venue starts with an initial ratio  $r_0$  slightly below the threshold  $\rho_2$ .  $P(r)$  will be negative, so  $\dot{r}$  will also be negative, and so the ratio begins to slide downwards. As it approaches  $r = 0$ , ever-smaller values of  $r$  cause the constant term  $-\varphi_2$  to dominate the polynomial, and the rate of decline  $\dot{r}$  becomes approximately proportional to  $1/r$ . A crash finally occurs because the integral converges to a finite value, meaning that the female proportion doesn’t just asymptotically fade away; it actually *hits zero in finite time*.

*Regarding Closed-form Solutions in the Time Domain*

We have seen that the trajectory of the population in the phase plane is determined by the integral relation between  $m$  and  $r$ :

$$\int \frac{dm}{m} = \int \frac{\mu_1 r^2 - \mu_2}{P(r)} dr \quad (23)$$

where  $P(r) = -\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2$  is a full cubic polynomial. In order to solve this integral analytically, we must perform a partial fraction decomposition, which in turn requires factoring the denominator into the form  $-\mu_1(r - \rho_1)(r - \rho_2)(r - \rho_3)$ . Though soluble in general, the symbolic expressions for these roots involve nested cube roots of complex discriminants (Cardano's formulae). Substituting these large symbolic expressions into the exponents of a power-law solution renders the result opaque and algebraically intractable. Without first quantifying the coefficients, we cannot know if the roots are real, complex, distinct, or repeated.

This, unfortunately, is but the first and most easily dealt-with impediment. Even in fortunate or contrived circumstances where we find coefficients that yield three clean, real roots (e.g.  $r \in \{1, 2, 3\}$ ), we are immediately met with a more formidable issue: inversion.

Let us assume, for the sake of argument, that we have found the trajectory  $m(r)$ . To find time evolution  $r(t)$ , we must solve the integral  $t(r) = \int \frac{m(s) \cdot s}{P(s)} ds$  (20). Solutions to this integral will invariably take the form of a sum of logarithmic and perhaps polynomial terms:

$$t(r) = A \log_e |r - \rho_1| + B \log_e |r - \rho_2| + C \log_e |r - \rho_3| + D. \quad (24)$$

To find explicit functions  $f(t)$  and  $m(t)$  we must invert this equation to find  $r$  as a function of  $t$ . Unfortunately an equation of the form  $t = \log_e(r) + \log_e(r - k)$  is not generally expressible in elementary closed form; typically only implicit forms or special functions are required. The solution  $r(t)$  notionally exists, but it will be a non-elementary implicit function (often involving the *Lambert W* function or worse for multiple logarithmic terms). Therefore, no general closed-form formula for  $f(t)$  can be given for this model. We

are forced to rely on parametric descriptions of the form  $(t(r), m(r), f(r))$ .

While the general case is intractable, there are specific constraints on the coefficients that simplify the algebraic treatment enough to allow for explicit solutions.

If we assume males are indifferent about the gender ratio when entering (as discussed previously) then  $\mu_1 = 0$ , and thus the term  $-\mu_1 r^3$  vanishes from  $P(r)$  which then becomes a quadratic equation  $P_2(r) = \varphi_1 r^2 + \mu_2 r - \varphi_2$ . Roots are then found easily via completion of the square or the quadratic formula. Integration leads to simpler inverse trigonometric or hyperbolic functions (such as *arctan* and *arctanh*) which can sometimes be successfully inverted to find  $r(t)$  explicitly.

If, on the other hand, we adopt a very ‘egalitarian’ standpoint where female and male sensitivities are identical ( $\varphi_1 = \mu_1$  and  $\varphi_2 = \mu_2$ ), the numerator and denominator in the phase trajectory will often share factors, which in turn means that ratio evolution  $\dot{r}$  might simplify to a constant or a simple power, potentially allowing for trivial solutions such as linear growth  $r(t) \propto t$ . This might be plausible in certain scenarios, such as when a female and a male attend as a couple, or in cultures where *Freikörperkultur* (FKK) is unremarkable and broadly accepted. (We shall return to this in another guise in due course.)

We have already analysed the intuitively very plausible “indifferent males” ( $\mu_2 = 0$ ) scenario, and though  $P(r)$  remains a cubic, the loss of the linear term  $r$  often simplifies the root structure. Hybridising the previous two simplifying cases, if females were to also become indifferent to being in stark minority ( $\varphi_2 = 0$ ) this would further simplify root structure, but observation and intuition suggest that  $\varphi_2$  is large and significant, probably due to expectations of obnoxious or even predatory male behaviour. We shall return to this in due course. Solutions would be parametric as the inversion problem would still exist.



*Forcing Viability: the “No Single Males” policy*

We have so far discussed “free entry” models, where the influx of females and males were independent functions of the current ratio  $r$ : with natural female influx per unit time given by

$$J_f^{nat} = \varphi_1 \frac{f}{m} = \varphi_1 r, \quad (25)$$

and natural male influx per unit time will be

$$J_m^{nat} = \mu_1 \frac{f}{m} = \mu_1 r. \quad (26)$$

To formalise the “no single males” policy, we model the venue as an optimiser facing a binding constraint. The policy dictates that female entry must equal or exceed male entry  $J_f \geq J_m$ . In the strictest or most draconian interpretation, we force  $\dot{m}_{in} \leq \dot{f}_{in}$ , thus creating a system with a binding constraint.

In this scenario of constrained dynamics, evolution of the male population is now governed by the minimum of “natural influx” and “policy limit”:

$$\dot{m} = \min(\mu_1 r, \varphi_1 r) - \frac{\mu_2}{r}. \quad (27)$$

Assuming the venue is popular enough that males want to enter at a higher rate than females do, (i.e.,  $\mu_1 > \varphi_1$  which is anecdotally not unreasonable and the root cause of the instability we modelled previously), the policy constraint becomes binding.

The system of ODEs that models the system collapse to is:

$$\begin{cases} \dot{f} = \varphi_1 \cdot r - \frac{\varphi_2}{r} \\ \dot{m} = \varphi_1 \cdot r - \frac{\mu_2}{r} \end{cases} \quad \text{with } \varphi_i, \mu_i, \in \mathbb{R}_{>0}. \quad (28)$$

(Note that in the expression for  $\dot{m}$   $\mu_1$  has been replaced by  $\varphi_1$ .)

Recall  $\dot{r} = \frac{1}{m}(\dot{f} - r\dot{m})$  from (9). Substituting in the constrained male rate (10) now becomes:

$$\dot{r} = \frac{1}{m} \left[ \left( \varphi_1 r - \frac{\varphi_2}{r} \right) - r \left( \varphi_1 r - \frac{\mu_2}{r} \right) \right] \quad (29)$$

which in turn simplifies to a modified version of (11), the equation that governs the overall gender ratio:

$$\dot{r} = \frac{1}{mr} (-\varphi_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2). \quad (30)$$

The original destabilising cubic term  $-\mu_1 r^3$  has been replaced by the ‘policy’ term  $-\varphi_1 r^3$ . As we generally assume  $\mu_1 > \varphi_1$  (males are more eager to frequent than females), replacing  $\mu_1$  with  $\varphi_1$  significantly flattens the downward curve of the cubic polynomial  $P_{policy}(r)$ . This pushes roots towards the left, effectively lowering the critical threshold required for viability. *Effectively, by tying male entry to female entry, policymakers artificially slow the rate of male growth to match the female ‘comfort’ growth rate and ensure the ratio is always viable by rationing male attendance.* The venue cannot become ‘oversaturated’ with males by the process of entry, only via differential exit (and as we have seen,  $\mu_2$  can be argued to be essentially zero).

*Shadow Costs: Unlocking the Value of Others*

In the unconstrained region (where the constraint is slack and not effective), the ‘cost’ of entry is zero (or the nominal access fee). However, when the constraint binds ( $\dot{m}_{in} = \dot{f}_{in}$  despite  $\mu_1 > \varphi_1$ ), we have *excess male demand* for admission.

There is now a difference between the number of males who wish to enter and the number who are granted access:

$$\Delta J_m = (\mu_1 - \varphi_1) \cdot r. \quad (31)$$

The *shadow price*  $\lambda$  represents the marginal value of a female entrant, as every woman who enters ‘unlocks’ a corresponding slot for a male, the presence of a female guest acquires measurable economic value equal to the profit that could be generated by that blocked male entrant.

We have already seen that unconstrained female demand flux is  $J_f^{dem} = \varphi_1 \cdot r$  and unconstrained male demand flux is  $J_m^{dem} = \mu_1 \cdot r$ , however due to policy constraints aimed at curtailing instability actual male entry  $J_m$  cannot exceed actual female entry  $J_f$ :

$$J_m \leq J_f. \quad (32)$$

In this “excess male demand” scenario, the venue is turning away males that are willing to pay in order to be granted access, depressing its revenue.

Since  $J_m$  is capped by  $J_f$ , total admitted flux is determined solely by female demand  $J_f$ . For every single female who chooses to enter, one further male is admitted. We can optimise this in two distinct manners.

Scenario A: Maximisation of Revenue ( $Z$ )

Consider the venue to be a commercial operator seeking to maximise total revenue  $Z$  (a proxy for profit and economic utility). The objective function is

$$Z \doteq P_f \cdot J_f + P_m \cdot J_m \quad (33)$$

where  $P_i$  represents access fees paid by each demographic to the venue.

To ensure the optimisation problem is well-posed and demographically meaningful, we must bound the domain. While our model still admits a venue having an infinite capacity, the market it draws from does not. Therefore we model the venue's market as a geographic Voronoi cell defined by travel-time or travel-cost distance relative to competing venues. Within this catchment area with finite population, there exists a maximum number of female and male guests ( $f_{max}$  and  $m_{max}$  respectively), and thus the optimisation occurs on the compact domain  $[0, f_{max}] \times [0, m_{max}]$ . Consequently, our objective function is continuous on a compact set, guaranteeing the existence of a global maximum by the Extreme Value Theorem, thereby discarding unphysical asymptotic solutions. We introduce the catchment constraints  $f \leq f_{max}$  and  $m \leq m_{max}$  and assign the multipliers  $\nu_f$  and  $\nu_m$ .

Under these constraints and the binding policy, the effective constrained revenue function becomes:

$$Z_{constrained} \doteq P_f \cdot J_f + P_m \cdot J_f = (P_f + P_m) \cdot J_f. \quad (34)$$

Under this scenario, the marginal value of one additional female guest is no longer just her access fee  $P_f$ , but rather it has now become

$$\frac{\partial Z}{\partial f_{in}} = P_f + P_m. \quad (35)$$

To rigorously ground this result, we invoke the Karush-Kuhn-Tucker (KKT) conditions (KARUSH, 1939; Kuhn and Tucker, 1951). Let  $\lambda \geq 0$  be the Lagrange multiplier associated with the policy constraint  $g(J_f, J_m) = J_m - J_f \leq 0$ . The Lagrangian for the

venue's maximisation problem is:

$$\mathcal{L}(J_f, J_m, \lambda, \nu_f, \nu_m) \doteq P_f J_f + P_m J_m - \lambda(J_m - J_f) - \nu_f(J_f - J_f^{max}) - \nu_m(J_m - J_m^{max}). \quad (36)$$

When the catchment constraints are slack, the Lagrangian collapses down to its operative form:

$$\mathcal{L}_{operative} = P_f J_f + P_m J_m - \lambda(J_m - J_f) \quad (37)$$

The first-order stationary condition with respect to female entry is (in the minimisation convention) is

$$\frac{\partial \mathcal{L}}{\partial J_f} = P_f + \lambda = 0 \quad (38)$$

or, more simply, the marginal value is increased by  $\lambda$ . When the constraint is binding (male demand for access is exceeding the quota), the shadow price  $\lambda$  exactly equals the marginal revenue of excluded males ( $P_m$ ), which should be subject to a highest-bidder auction in order to preferentially satisfy those who most desire to attend. Consequentially, the effective marginal value of a female guest becomes  $V_f = P_f + \max(P_m)$ . This multiplier  $\lambda$  represents the formal economic justification for subsidies to support female attendance such as the schemes described in (Pollen, 2021, 2022): the subsidy creates a positive externality every time an additional female is enticed to attend, and to maximise total surplus, it must internalise this externality by subsidising her entry up to the value  $\lambda$ . The male premium pays for the female subsidy to relax the binding constraint.

Scenario B: Maximisation of Attendance ( $W$ )

Contrast this with a community-run venue seeking to maximise total participation and welfare  $W$  of its attendees, approximated here by total attendance

$$W \doteq J_f + J_m. \quad (39)$$

The optimisation problem is distinct, and the binding constraint yields the result

$$W_{constrained} \doteq 2J_f. \quad (40)$$

To maximise attendance and welfare, the venue must also maximise female attendance  $J_f$  as each additional female *still* unlocks a corresponding male vacancy. Thus we arrive at a powerful conclusion: under the constraints of the “no single male” policy, the incentives of revenue-maximisation and welfare-maximisation are perfectly aligned. In both scenarios, the optimal strategy is invariant: the female guest is the limiting resource, and just as her preferences  $\varphi_1, \varphi_2$  govern her propensity to attend, her shadow price governs the potential scale of the venue’s overall attendance and acquires value as the finite resource that unlocks revenue and attendance in a context of induced scarcity.

*Trading Off: Marginal Rates of Substitution*

Marginal Rate of Substitution (MRS) describes and quantifies the trade-off required to maintain a constant level of “social utility” (or growth stability) at the venue.

Let us define the utility of females for the venue  $U_f$  as equivalent to the female growth rate (a proxy for comfort/attractiveness of the venue):

$$U_f(m, f) \doteq \dot{f} = \varphi_1 \frac{f}{m} - \varphi_2 \frac{m}{f} \quad (41)$$

The MRS is thus the answer to the question: “if we add one male (increasing discomfort), how many females must we add to keep the overall comfort level unchanged?”

The slope of the iso-comfort curve  $dU_f = 0$  is given by:

$$\frac{\partial U_f}{\partial f} + \frac{\partial U_f}{\partial m} = 0. \quad (42)$$

Taking the partial derivatives results in:

$$\frac{\partial U_f}{\partial f} = \frac{\varphi_1}{m} + \frac{\varphi_2 m}{f^2}, \quad \frac{\partial U_f}{\partial m} = \frac{\varphi_1 f}{m^2} - \frac{\varphi_2}{f}. \quad (43)$$

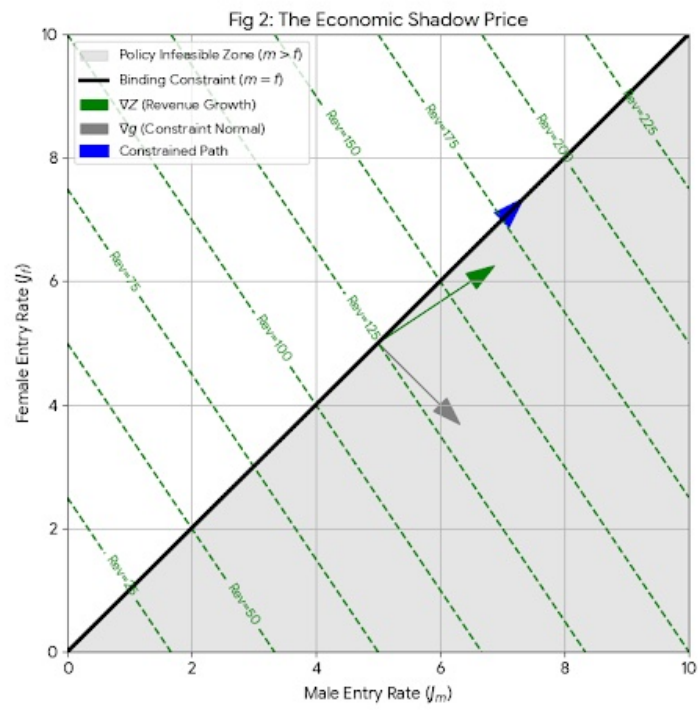
Solving for the gradient  $\frac{df}{dm}$  we find:

$$MRS_{m,f} = -\frac{df}{dm} = \frac{-\left(-\frac{\varphi_1 f}{m^2} - \frac{\varphi_2}{f}\right)}{\frac{\varphi_1}{m} + \frac{\varphi_2 m}{f^2}}. \quad (44)$$

Multiplying both numerator and denominator by  $m$  gives:

$$MRS_{m,f} = \frac{\varphi_1 r + \varphi_2 / r}{\varphi_1 + \varphi_2 / r^2} = r \cdot \left( \frac{\varphi_1 r^2 + \varphi_2}{\varphi_1 r^2 + \varphi_2} \right) = r, \text{ and thus } MRS = \frac{f}{m}. \quad (45)$$

In demographic terms, this means that in order to maintain the exact same level of comfort, the substitution rate is simply the current ratio. This both confirms and follows from isoclines (lines of constant growth) being rays from the origin. To keep the comfort constant, the venue must expand population linearly along the ray.





### *Full House: Finite Carrying Capacity*

So far, our model has assumed unconstrained growth. However, as every physical venue has a finite carrying capacity  $K$  (determined, amongst other things, by locker counts, sun-beds, limited accommodation or camping pitches, pool safety limits, or fire codes). Unlike biological and demographical systems where growth slows due to resource scarcity (the logistic damper of Verhulst), a venue usually hits a hard constraint when whatever limiting resource is fully subscribed. Furthermore, as potential attendees are neither being born nor dying according to the abundance of resources, but simply turned away when  $K$  is reached, it seems preferable to model a “turnstile model”<sup>2</sup>: not a damping term on the parameters, but a boundary condition on the domain. The system evolves according to its standard equations until the total population  $f + m = K$ . We can formalise this as a constraint on the total number of attendees:

$$f + m \leq K. \tag{46}$$

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2. There is an additional issue of information asymmetry to be taken into account: naturist venues, by their very nature, tend to be private and not easily viewable from outside. This entails that prospective entrants presenting themselves at the gate will be unaware of the current level of attendance or crowding inside the venue’s borders. Consequentially only gatekeepers who are aware of the venue’s internal rate of occupancy are in a position to moderate attendance accordingly.

Dynamics at Capacity: the “Churn Mechanism”

Once the venue is full and the constraint

$$f + m = K \quad (47)$$

holds, dynamics switch from growth to replacement. We must also be careful to maintain the demographic balance between  $f$  and  $m$  in order to avoid a “death spiral” by ensuring demographics must settle on the stable attractor ray

$$f = r^* m. \quad (48)$$

We are going to solve for the point that satisfies both conditions simultaneously. Substituting the stable ratio constraint (48) into the capacity constraint (47) results in:

$$m + r^* m = K \quad (49)$$

whereupon we can factor out  $m$  to obtain

$$m(1 + r^*) = K. \quad (50)$$

Now we can solve for the male equilibrium population  $m^*$ :

$$m^* = \frac{K}{1 + r^*} \quad (51)$$

and then substitute back into (44) to find the female equilibrium

$$f^* = \frac{K r^*}{1 + r^*}. \quad (52)$$

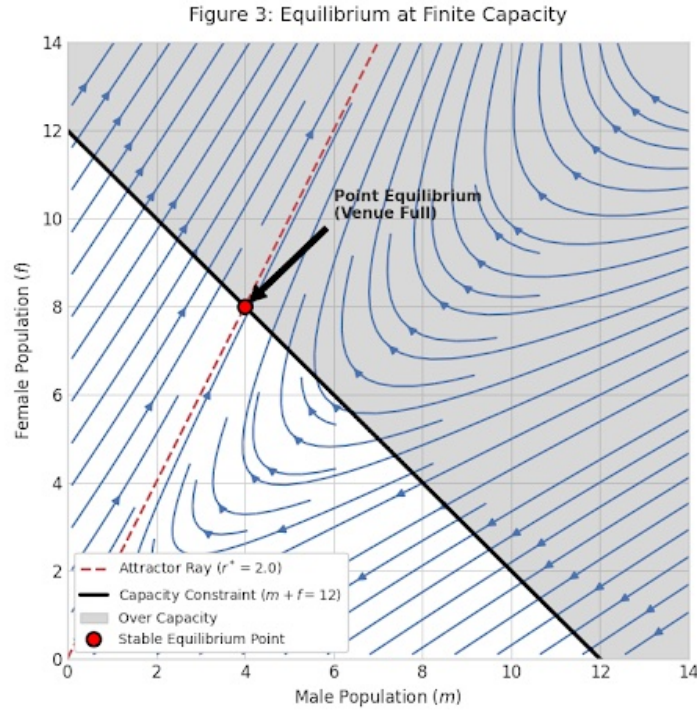
This point  $(f^*, m^*)$  represents the only sustainable “full occupancy” scenario. If a venue reaches capacity at a different ratio, the system is in one of two conditions:

If  $r > r^*$  the venue has a female-heavy demographic. The venue is stable and saturated but admissible males cannot gain entry. If the venue differentiates price by gender and subsidises female participation, it is leaving revenue on the table.

If  $r < r^*$  the venue has a male-heavy demographic and is in the unstable ‘repulsor’ zone. In a regime of mere replacement, the system cannot auto-correct: suppose the venue hits capacity with a male-dominated ratio  $r < r^*$ . Males, having a lower sensitivity to gender imbalance

$\mu_2 \approx 0$  are unlikely to leave. However, females, feeling crowded out and in discomfort, will exit ( $\dot{f}_{out} > 0$ ). These female exits create new availability for new entrants which by definition will have to be females, because at capacity any vacancy created by a female is going to be reserved for the entry of a new female guest. However, replacing dissatisfied females with other dissatisfied females does not restore equilibrium but maintains the  $r < r^*$  regime.

In order to avoid this, the venue must most likely run at  $f + m < K$  (below capacity), sacrificing immediate revenue in order to prevent irrecoverable female population collapse.



*When women are at ease*

We have stated that it is natural to assume that  $\varphi_2 \gg 0$  because, anecdotally, females are made uneasy by expectations of inappropriate behaviour. We have also stated that when reassured by male companionship,  $\varphi_2 \approx \mu_2 \approx 0$ . For economy of argument, let us now return to the free-entry model. Recall that gender ratio dynamics are governed by the cubic (9):

$$P(r) = -\mu_1 r^3 + \varphi r^2 + \mu_2 r - \varphi_2 \quad (53)$$

with equilibria at  $P(r^*) = 0$ . If we implicitly differentiate the root, we find that

$$P(r_i, \varphi_2) = 0 \iff \frac{dr_i}{d\varphi_2} = -\frac{\partial P / \partial \varphi_2}{P'(r_i)} = \frac{1}{P'(r_i)}, \quad (54)$$

which we can interpret this thusly:

At the unstable root  $r_1$ , we have our repulsor  $P'(r_1) > 0$ , so  $\frac{r_1}{d\varphi_2} > 0$ , meaning that the unstable root moves to the right as  $\varphi_2$  increases;

At the stable root  $r_2$ , we have our attractor  $P'(r_2) < 0$ , so  $\frac{r_r}{d\varphi_2} < 0$ , meaning that the stable root moves to the left as  $\varphi_2$  increases.

This has a remarkable consequence: a rising  $\varphi_2$  causes the two positive roots to move towards each-other until they coalesce into a saddle-node at the boundary case where the local maximum of  $P$  is exactly zero. When  $\varphi_2 = 0$  we have that for  $P(0) = 0$  and small  $r > 0$  we have that

$$P(r) \approx \mu_2 r + \varphi_1 r^2 > 0, \quad (55)$$

which means that in these circumstances  $\dot{r} > 0$  and thus the *ratio  $r$  is pushed upwards rather than sucked into collapse* and indeed it is evident

$$\varphi_2 = 0 \iff P(r) = -\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r = r(-\mu_1 r^2 + \varphi_1 r + \mu_2) \quad (56)$$

as the constant term disappears.

Thus we have that for small  $r > 0$   $P(r) \approx \mu_2 r > 0$ ,  $\dot{r} > 0$  and we can interpret this

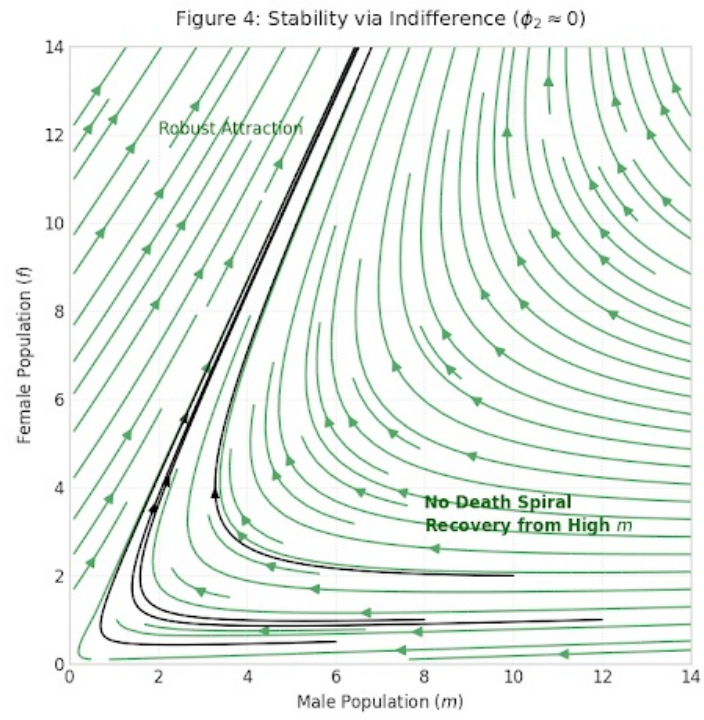
as an impediment to triggering the “death spiral” discussed previously. Mathematically, this breaks homogeneity because now there is a root at  $r = 0$  and the sign flips.

Demographically, this frees the community from the “tyranny of the origin” whereby every  $r^*$  must pass through the origin. This explains how and why we very frequently observe that small, high-trust groups (such as, for example, four or five heterosexual couples) at a venue can easily accommodate a ratio-breaking event such as the arrival of a single male or a male homosexual couple. This is fully compatible with the compact proof of stability presented earlier.

$\varphi_2$  is a coefficient associated with female tolerance for being in the minority, but it is likely first and foremost a measure of justified female concern with inappropriate male behaviour (because otherwise they wouldn’t even be considering attending a naturist venue). This is likely why couples exhibit  $\varphi_2 \approx \mu_2 \approx 0$  behaviour: because the female feels safe from inopportune attention. If a norm of male propriety could be strictly enforced, the system can literally “get off to a good start” where high trust and ease create an environment that is receptive to stable growth and far more robust to imbalance than we had heretofore considered<sup>3</sup>.

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3. This is not to be confused with the “dicy venue” third case of a repeated positive root, and is in many ways its opposite: while that was a case of chronic instability we can assume to be so transient as to never observe in the field, these are circumstances in which imbalance is readily tolerated because the system is building upon a firm foundation of mutual trust.



### The Geometry of Safety

To understand why the specific constraint  $r \leq 1$  is so frequently advocated, we must look beyond the aggregate demographics and examine the detailed geometry of the venue. The ‘comfort’ of a female guest, and specifically her ability to remain indifferent to the number of males present ( $\varphi_2 \approx 0$ ) is not merely a psychological trait, but an indicator of her local connectivity to other attendees.

We can model the venue floor as a stochastic Voronoi tessellation. A “safe space” for a female guest can be defined topologically: she feels secure if she is part of a couple or if she belongs to a connected component of other female guests large enough to form a “social shield” against isolation or encirclement. This maps the problem to percolation theory.

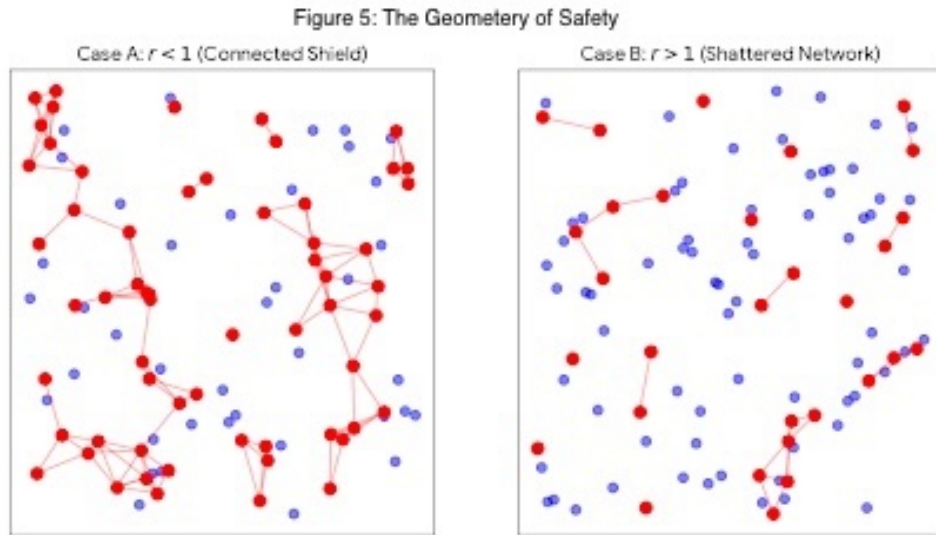
It is a known result in stochastic geometry that the critical probability threshold  $p_c$  for the emergence of a “giant component” (a connected subgraph) on a Voronoi tessellation is  $p_c \approx 1/2$ . If  $p_f$  represents the probability that a random guest is female, a robust, connected female network only exists if  $p_f \geq 1/2$ . Expressing this in terms of our ratio  $r = f/m$  we have:

$$p_f = \frac{f}{f+m} = \frac{1}{1+1/r} \iff 1 \geq \frac{1}{2} + \frac{1/2}{r} \iff r \leq 1 \quad (57)$$

This result provides the physical justification for the behavioural parameters used earlier in this paper: the assumption that females can be “at ease” and tolerant of a prevalence of male strangers is valid as long as this giant component is maintained. The constraint  $r \leq 1$  preserves the topological integrity of the female social graph in conditions of uncertainty. Above this ratio ( $r > 1$ ) the female network undergoes a phase-transition and shatters into isolated, vulnerable, encircled clusters, likely reactivating the sensitivity parameter  $\varphi_2$  and triggering defensive behaviour.

These two results are not contradictory: combined, they explain why amongst strangers, women prefer a ratio that does not disfavour them heavily, while when assembled

in a group they can form a stable defensive network. Both mechanisms yield the  $\varphi_2 \approx 0$  value that is crucial in avoiding female population collapse in situations of unfavourable gender ratios.





## *Conclusion*

This paper presented a dynamical systems approach to modelling the demographics of naturist venues. By quantifying gender-specific preferences for, and sensitivities to, population ratios, we derived an equation (11) that governs evolution of the venue’s social balance.

Our analysis yields three principal theoretical conclusions:

### 1. Inherent Instability

The proverbial “invisible hand” does not guarantee a balanced gender ratio and to some extent can punch a venue in the groin. We identified the existence of a critical unstable root below which the venue enters a “death spiral” of male-dominated stagnation. This indicates that passive management is insufficient because a venue cannot organically “grow its way out” of a bad initial ratio if our intuitions about typical coefficients are correct ( $\varphi_2 \gg 0, \mu_2 \approx 0$ ).

### 2. Shadow Price of Balance

By modelling the “no single males” policy as a binding constraint, we derived KKT conditions for the system. We demonstrated that under this constraint, the shadow price of a female entrant includes the opportunity cost of the male entrants she enables. This provides a rigorous economic justification for cross-subsidisation (e.g., “ladies free” pricing), showing that revenue maximisation ( $Z$ ) (as a proxy for economic utility) and community participation maximisation ( $W$ ) effectively converge on the same strategy: prioritising female attendance and participation.

### 3. The Determinism of Capacity

We showed that when a venue reaches its physical carrying capacity  $K$ , the final equilibrium  $(f^*, m^*)$  is strictly determined by the intersection of the stable gender ratio and the carrying capacity border.

This model suggests that a naturist venue is not merely a static facility but a complex social reactor. It suggests that care must be exercised when priming a naturist venue with its first participants, and that seeding the initial population with a selection of females who are largely indifferent to being outnumbered by males  $\varphi_2 \approx 0$  could go a long way towards avoiding the scenarios of collapse the model predicts when imbalance sets in beneath the critical ratio that would ordinarily trigger the “death spiral” process.

*On the “no single male” policy: motivation and consequences*

Some remarks about the “no single male” policy are in order, as it is notoriously contentious. Given the model and the assumptions about coefficients, it is shown to be an effective policy to prevent collapse of the female population from which there is no endogenous return. Whether a venue chooses to adopt it or not is largely down to their values and whether the preferences of their prospective female guests (as described in aggregate by their coefficients) make the prospect of female attendee collapse a realistic scenario which warrants guarding against. As noted above, even a small “hard-core” kernel of females unconcerned by males significantly outnumbering them can provide the bedrock that in turn could allow slightly more sensitive, but nonetheless less sensitive than average, females to participate. While the mathematical model emphasises the primacy of female preferences and behaviours, it is likely that male behaviour and attitudes are a determining factor in how at ease females feel at venues and therefore how inclined they are to stay as opposed to depart when gender ratios are unbalanced in their disfavour. When in heterosexual couples, both partners empirically display a general indifference to gender imbalance ( $\varphi_2 \approx \mu_2 \approx 0$ ), presumably because the female feels protected. We must also keep in mind that coefficients are aggregate values, and that if there are enough females with  $\varphi_2 \approx 0$ , the community average can decrease to the point that the unstable root disappears entirely, making the “death spiral” scenario impossible.

The “no single male” rule is not a fundamental requirement for safety; it is an engineering control that compensates for insufficient trust and weak norm enforcement. In high-trust settings, women’s discomfort parameter effectively vanishes, and the venue can also tolerate male-skewed ratios provided women are not locally isolated or encircled. In low-trust settings, the same local-connectivity condition fails more easily, so the rule appears necessary because it preserves the female “giant component” (social shielding) by restricting

male influx. We have also proven that when surrounded by a close-knit community they trust, females can accomodate growth that disfavours them demographically provided it does not become overwhelming. This is presumably the process by which naturist communities take root, mature, and grow: by beginning with trust and being built with mutal respect.

### Stepping back

In closing, the authors would like to make a final remark: if the reader is uncomfortable with the rational economic consequences that derive from the “no single male” policy as described in this paper, *in primis* the emergence of a rational incentive to promote and incentivise female participation through asymmetric pricing that makes females themselves “objects of pursuit”, then the reader cannot consistently advocate in favour of that policy. Though never explicit, “no single males” introduces a form of rationing and artificial scarcity which objectifies female participants by giving their presence economic value: stability is introduced, but at the great social cost of making females a precious finite resource. This should cause us to pause and reflect. Perhaps, as alluded to in the previous paragraph, the way forward is to adopt norms of male behaviour that place female participants at ease and adhere to them strictly. Making females’ participation a commodity to be bought is, surely, not in the interests of the broader naturist community to whom such a consequence should be antithetical to the point of rejecting the premise.

## Reference List

- Becker, G.S. (1957) *The economics of discrimination: an economic view of racial discrimination*. University of Chicago.
- Bollob'as, B. and Riordan, O. (2006) *Percolation*. Cambridge University Press.
- Broadbent, S.R. and Hammersley, J.M. (1957) "Percolation processes," *Mathematical Proceedings of the Cambridge Philosophical Society*, 53(3), pp. 629–641. doi:10.1017/s0305004100032680.
- James, B. (1965) "An economic theory of clubs," *Economica*, 32(125), pp. 1–14.
- KARUSH, W. (1939) "Minima of functions of several variables with inequalities as side constraints," *M. Sc. Dissertation. Dept. of Mathematics, Univ. of Chicago* [Preprint]. Available at: <https://cir.nii.ac.jp/crid/1572824501070501888>.
- Kuhn, H. and Tucker, A. (1951) "Proceedings of 2nd berkeley symposium," in *Proc. 2nd Berkeley Symp.*, pp. 481–492.
- Pollen, A. (2021) *Nudism in a Cold Climate The Visual Culture of Naturists in Mid-20th Century Britain*. 1st edn. Atelier Editions.
- Pollen, A. (2022) "Cold Comfort?," *H&E Naturist*, pp. 62–65.
- Schelling, T.C. (1971) "Dynamic models of segregation†," *The Journal of Mathematical Sociology*, 1(2), pp. 143–186. doi:10.1080/0022250x.1971.9989794.
- Voronoi, G. (1908) "Nouvelles applications des paramètres continus à la théorie des formes quadratiques. Deuxième mémoire. Recherches sur les paralléloèdres primitifs.," *Journal für die reine und angewandte Mathematik (Crelles Journal)*, 1908(134), pp. 198–287. doi:10.1515/crll.1908.134.198.

## Appendix

Though the model is explicitly a toy for coarsely modelling phenomena, a key question for empirical verification is whether the theoretical coefficients  $(\varphi_1, \varphi_2, \mu_1, \mu_2)$  can be recovered from real-world data collected in the field. While a single observation of a venue's gender balance (observing only  $r$  or more likely  $r^*$  if in equilibrium) presents an identification problem because an infinite set of parameter combinations can yield the same root, a time-series of attendance provides sufficient variation to estimate coefficients. Recall original model formulated in (1) and (2).

For a discrete time-series (taken at hourly or daily intervals) we approximate the instantaneous rate of a gender  $\dot{g}_{f \vee m}$  as the finite difference

$$\Delta g_{t,f \vee g} = g_{t+1,f \vee g} - g_{t,f \vee g}, \quad (1)$$

that can be rearranged into a standard linear regression model of the form

$$Y = \beta_1 X_1 + \beta_2 + \epsilon. \quad (2)$$

### Estimating female coefficients

Let the dependent variable be the net change in females:

$$Y_{f,t} = f_{t+1} - f_t \quad (3)$$

and let the independent variables be the current ratios:

$$X_{1,t} = \frac{f_t}{m_t}, X_{2,t} = \frac{m_t}{f_t}. \quad (4)$$

We estimate the regression

$$Y_{f,t} = \varphi_1 X_{1,t} - \varphi_2 X_{2,t} + \epsilon, \quad (5)$$

where the regression coefficient for  $X_{1,t}$  gives the estimate for  $\varphi_1$  and the coefficient for  $X_{2,t}$  gives the estimate for  $-\varphi_2$ .

### Estimating male coefficients

Similarly, let

$$Y_{m,t} = m_{t+1} - m_t. \quad (6)$$

We estimate

$$Y_{m,t} = \mu_1 X_{1,t} - \mu_2 X_{2,t} + \epsilon. \quad (7)$$

### Data Requirements

To obtain statistically significant estimates, the data must contain **variance in the ratio  $r$** . If, on the other hand, a venue is perfectly managed and always sits at equilibrium  $r^* = f^*/r^*$ , the regressors  $X_1$  and  $X_2$  will be unchanging constants. Similarly if observations are made in situations where there is no influx or efflux, such as on a nude cruise. In these cases regression will suffer from perfect collinearity, rendering the coefficients unidentified.

Valuable data is generated when the system is out of equilibrium. An “event study” approach analysing days when the ratio is eternally perturbed (e.g., a bachelor or bachelorette party arrives or departs) provides a useful exogenous shock to trace out response curves ( $X$  values) and the system’s correction ( $Y$  values).

### Implied Structural Tests

Once the coefficients are estimated from the data, the theoretical “Survival Inequality” derived from this paper can be empirically tested. Though derived from what is explicitly a toy model amenable to analysis, it would be nonetheless interesting to ascertain to what degree this inequality qualitatively and quantitatively describes actual observations in the field. If viable venues’ guests’ estimated parameters are found to be broadly compatible with those prescribed by the characteristic polynomial  $P(r)$ , this would justify it as being a potential object of study and social engineering to ensure robustness.